

AgriRevise Development Progress Logbook

Mobile-first educational revision website with database leaderboard, game progression, scoring, and responsive viewport improvements.

Prepared for supervisor review

Generated: 16 May 2026

Frontend UI

MySQL/PHP API

Leaderboard

Game Logic

Mobile Viewport

Mula

1. Project Summary

AgriRevise is a web-based agriculture revision tool designed around short interactive games, topic progression, and a score leaderboard. The most recent development work focused on connecting player scores to a MySQL database, improving the leaderboard, locking game progression, and refining all mobile game pages so they fit properly in a single phone viewport.

Main outcome

Player names and scores are now prepared for persistent database storage and leaderboard ranking.

User flow

A player enters a name, plays games, keeps the best score per game, and appears in Papan Prestasi.

Progression

Game cards can be locked until prerequisite activities are completed.

Responsive goal

Main game screens were adjusted to use the available viewport without overflowing or leaving large empty areas.

2. Start-to-Finish Progress Log

Phase	Work Completed	Evidence / Result
Database planning	Designed a MySQL players table with ID, case-insensitive name key, individual game score columns, and total score.	Table supports one row per player name and leaderboard sorting by total or game-specific score.
Backend API	Added PHP endpoints for player creation, score saving, and leaderboard retrieval.	Scores update through the API and total score is recalculated from all game columns.
Highest score logic	Implemented local and server-side best-score preservation.	If a player gets 13/15, then later 14/15, the stored score becomes 14/15. Lower retries do not overwrite higher scores.
Leaderboard UI	Updated Papan Prestasi to show 5 players per page, compact rows, sort selector, and top pagination controls.	Leaderboard now supports total score and individual game score views.
Game locking	Connected locked topic/game cards to completed-game local storage flags.	Pengiraan Kos Baja is locked until Jenis Baja is completed.
Mobile viewport pass	Reviewed and adjusted all main game pages for phone-sized layout.	Cards, text, answer boxes, drag/drop zones, and calculator keypad now use the available viewport more evenly.
Final polish	Improved the cash mission truck animation and purchase order board.	Truck now enters from the left, waits, receives the packet, then exits right after order approval.

3. Database and Scoring Implementation

The database design uses a single **players** table. The **name_key** column stores a normalized version of the player name, so the same player can continue using old scores by logging in with the same name.

database/create_players_table.sql

```
CREATE TABLE IF NOT EXISTS players (
  id INT UNSIGNED NOT NULL AUTO_INCREMENT,
  name VARCHAR(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT NULL,
  name_key VARCHAR(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT NULL,

  score_jenis_sifat_tanah TINYINT UNSIGNED NOT NULL DEFAULT 0,
  score_tanah_tapak_penanaman TINYINT UNSIGNED NOT NULL DEFAULT 0,
  score_struktur_tanah TINYINT UNSIGNED NOT NULL DEFAULT 0,
  score_ph_tanah TINYINT UNSIGNED NOT NULL DEFAULT 0,
  score_kaedah_memperbaiki_tanah TINYINT UNSIGNED NOT NULL DEFAULT 0,
  score_jenis_baja TINYINT UNSIGNED NOT NULL DEFAULT 0,
  score_pengiraan_kos_baja TINYINT UNSIGNED NOT NULL DEFAULT 0,

  total_score SMALLINT UNSIGNED NOT NULL DEFAULT 0,
  PRIMARY KEY (id),
  UNIQUE KEY uq_players_name_key (name_key),
  KEY idx_players_total_score (total_score)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

The API uses **GREATEST()** to preserve the best score in MySQL. This avoids losing a previous high score when a player repeats a game and scores lower.

api/db.php - highest score update

```
$statement = $pdo->prepare(
  'UPDATE players
   SET `` . $column . `` = GREATEST(`` . $column . `` , :score),
      updated_at = CURRENT_TIMESTAMP
   WHERE id = :id'
);

$statement->execute(array(
  ':score' => $score,
  ':id' => (int) $player['id'],
));

agrirevisi_update_total_score($pdo, $player['id']);
```

score-api.js - local best score and API save

```
function setLocalGameScore(gameKey, score) {
  const game = games[gameKey];
  const cleanScore = Math.max(0, Math.min(game.maxScore, Math.trunc(Number(score) || 0)));
  const bestScore = Math.max(getLocalGameScore(gameKey), cleanScore);

  localStorage.setItem(game.scoreKey, String(bestScore));
  localStorage.setItem(game.completedKey, 'true');
  return bestScore;
}

const localBest = setLocalGameScore(key, score);
const payload = await request('score.php', {
  method: 'POST',
  body: JSON.stringify({
    name: playerName,
    game: key,
    score: localBest
  })
});
```

4. Leaderboard and Progression Features

The leaderboard page can sort by total score or by a specific game. It displays five players per page and provides navigation controls at the top of the board.

leaderboard.js - five players per page

```
let currentSort = sortSelect ? sortSelect.value : 'total';
let currentPage = 1;
const pageSize = 5;

function getSortedPlayers() {
  return players
    .slice()
    .sort((a, b) => getScore(b) - getScore(a) ||
      Number(b.total_score || 0) - Number(a.total_score || 0));
}

const visiblePlayers = getSortedPlayers()
  .slice((currentPage - 1) * pageSize, currentPage * pageSize);
```

script.js - locked game card logic

```
setupLockedCard({
  cardId: 'kos-baja-card',
  overlayId: 'kos-baja-lock-overlay',
  storageKey: 'jenisBajaCompleted',
  message: 'Selesaikan permainan "Jenis Baja" terlebih dahulu.'
});

if (gameUnlocked !== 'true') {
  card.classList.add('locked');
  card.addEventListener('click', (e) => {
    e.preventDefault();
    alert(message);
  });
}
```

5. Mobile Viewport and Game UI Improvements

The mobile design work focused on making each game feel complete within one phone screen. Instead of only increasing the outer box height, the internal elements were also resized so the text, buttons, drop zones, and visual assets remain balanced.

jenis-tanah-games.css - station layout uses full viewport

```
.tdtp-game-body .tdtp-game-container {
  flex: 1 1 auto;
  min-height: 0;
  display: flex;
  flex-direction: column;
}

.tdtp-game-body .question-card {
  flex: 1 1 auto;
  min-height: 0;
  padding: clamp(12px, 2.2dvh, 18px) 12px;
  display: flex;
  flex-direction: column;
}

.tdtp-game-body .drop-zones {
  flex: 1 1 auto;
  grid-template-rows: repeat(3, minmax(0, 1fr));
}
```

pengiraan-kos-baja.css - truck animation

```
@keyframes jbLorryArrive {
  0% { opacity: 0; transform: translateX(-82px) scale(0.58); }
  74% { transform: translateX(6px) scale(0.58); }
  100% { opacity: 1; transform: translateX(0) scale(0.58); }
}

@keyframes jbLorryDeliver {
  0%, 18% { transform: translateX(0) scale(0.58); }
  32% { transform: translateX(10px) scale(0.58); }
  100% { transform: translateX(360px) scale(0.58); }
}
```

6. Code Snippet Screenshots

These screenshots show the main implementation areas: MySQL table creation, best-score saving, leaderboard pagination, and the calculator game truck animation.

database/create_players_table.sql

```
1: CREATE TABLE IF NOT EXISTS players (
2:   id INT UNSIGNED NOT NULL AUTO_INCREMENT,
3:   name VARCHAR(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT NULL,
4:   name_key VARCHAR(100) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT NULL,
5:
6:   score_jenis_sifat_tanah TINYINT UNSIGNED NOT NULL DEFAULT 0,
7:   score_tanah_tapak_penanaman TINYINT UNSIGNED NOT NULL DEFAULT 0,
8:   score_struktur_tanah TINYINT UNSIGNED NOT NULL DEFAULT 0,
9:   score_ph_tanah TINYINT UNSIGNED NOT NULL DEFAULT 0,
10:  score_kandah_memperbaiki_tanah TINYINT UNSIGNED NOT NULL DEFAULT 0,
11:  score_jenis_baja TINYINT UNSIGNED NOT NULL DEFAULT 0,
12:  score_pengiraan_kos_baja TINYINT UNSIGNED NOT NULL DEFAULT 0,
13:
14:  total_score SMALLINT UNSIGNED NOT NULL DEFAULT 0,
15:  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
16:  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
17:
18:  PRIMARY KEY (id),
19:  UNIQUE KEY uq_players_name_key (name_key),
20:  KEY idx_players_total_score (total_score),
21:  KEY idx_players_score_jenis_sifat_tanah (score_jenis_sifat_tanah),
22:  KEY idx_players_score_tanah_tapak_penanaman (score_tanah_tapak_penanaman),
23:  KEY idx_players_score_struktur_tanah (score_struktur_tanah),
24:  KEY idx_players_score_ph_tanah (score_ph_tanah),
25:  KEY idx_players_score_kaedah_memperbaiki_tanah (score_kaedah_memperbaiki_tanah),
26:  KEY idx_players_score_jenis_baja (score_jenis_baja),
27:  KEY idx_players_score_pengiraan_kos_baja (score_pengiraan_kos_baja)
28: ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

MySQL players table with case-insensitive name key and score columns.

api/db.php - keep highest score and total

```
221: function agrirevise_update_total_score($pdo, $playerId)
222: {
223:     $sql = 'UPDATE players SET total_score = ' . implode(' + ', agrirevise_score_columns()) . ' WHERE id =
224:     :id';
225:     $statement = $pdo->prepare($sql);
226:     $statement->execute(array(':id' => (int) $playerId));
227: }
228: function agrirevise_save_score($pdo, $name, $game, $score)
229: {
230:     $meta = agrirevise_game_meta($game);
231:
232:     if (!$meta) {
233:         agrirevise_error('Unknown game key.');
```

PHP API keeps the highest score and recalculates total score.

leaderboard.js - five-player pagination

```
73: function getTotalPages() {
74:     return Math.max(1, Math.ceil(players.length / pageSize));
75: }
76:
77: function updatePagination() {
78:     const totalPages = getTotalPages();
79:
80:     currentPage = Math.min(Math.max(currentPage, 1), totalPages);
81:
82:     if (pageLabel) {
83:         pageLabel.textContent = `${currentPage} / ${totalPages}`;
84:     }
85:
86:     if (prevBtn) {
87:         prevBtn.disabled = currentPage <= 1;
88:     }
89:
90:     if (nextBtn) {
91:         nextBtn.disabled = currentPage >= totalPages;
92:     }
93:
94:     if (pagination) {
95:         pagination.classList.toggle('hidden', players.length === 0);
96:     }
97: }
98:
99: function renderLeaderboard() {
100:  updateHeading();
101:  leaderboardList.innerHTML = '';
102:  updatePagination();
103:
104:  if (!players.length) {
105:      setStatus('Tiada skor lagi.');
```

Leaderboard pagination displays five players per page.

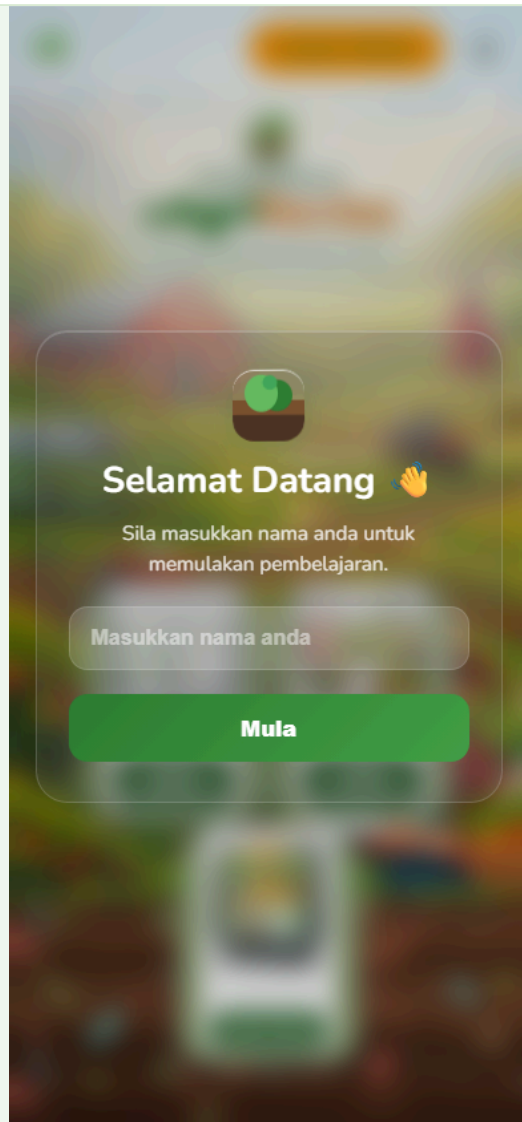
pengiraan-kos-baja.css - lorry animation

```
811: .jb-lorry {
812:   right: 10px;
813:   bottom: 65px;
814:   width: 100px;
815:   height: 62px;
816: }
817:
818: .jb-scene-warehouse .jb-lorry {
819:   left: 9px;
820:   right: auto;
821:   bottom: 65px;
822:   width: 100px;
823:   height: 62px;
824:   opacity: 1;
825:   transform: translateX(0) scale(0.58);
826:   transform-origin: left bottom;
827: }
828:
829: .jb-scene-warehouse .jb-lorry-arrive .jb-lorry {
830:   animation: jbLorryArrive 0.78s cubic-bezier(0.2, 0.84, 0.28, 1) both;
831: }
832:
833: .jb-lorry-box {
834:   position: absolute;
835:   left: 0;
836:   bottom: 14px;
837:   width: 72px;
838:   height: 39px;
839:   border-radius: 8px 4px 4px 8px;
840:   background: linear-gradient(180deg, #f7f3e2, #d7be7c);
841:   border: 2px solid rgba(126, 92, 38, 0.28);
842: }
843:
844: .jb-scene-warehouse .jb-lorry-box {
845:   left: 0;
846:   bottom: 14px;
847:   width: 72px;
848:   height: 39px;
849:   border: 2px solid rgba(126, 92, 38, 0.28);
850:   border-radius: 8px 4px 4px 8px;
851:   background: linear-gradient(180deg, #f7f3e2, #d7be7c);
852: }
853:
854: .jb-scene-warehouse .jb-lorry-box::after {
855:   content: "AGRIREVISE";
```

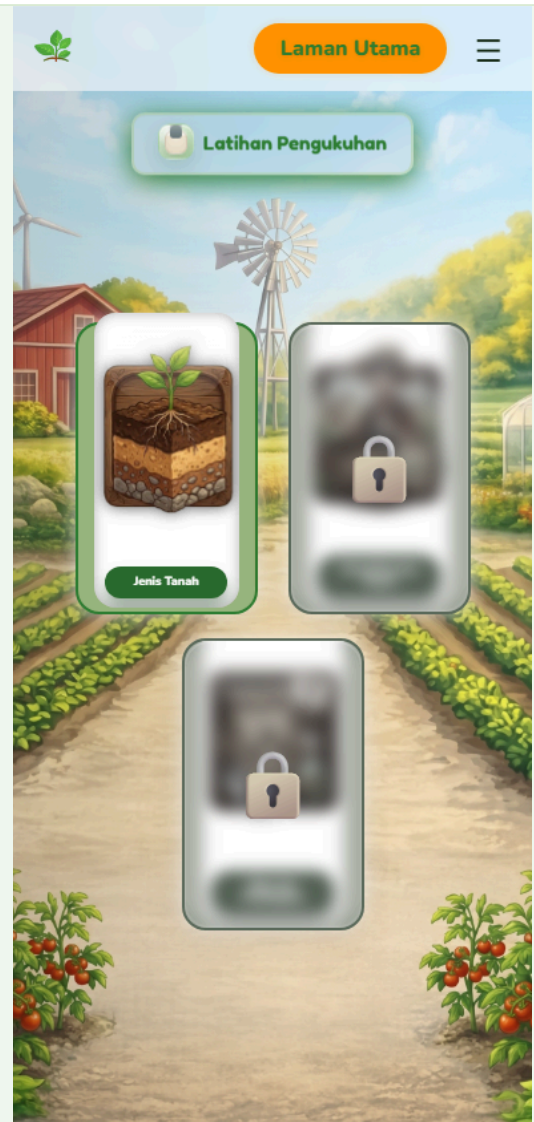
Calculator game animation and purchase order styling.

7. Website Design Screenshots

Screenshots were captured from the current local project using a 390 x 844 mobile viewport. The leaderboard image uses mock data only for visual demonstration because local file preview cannot call the Hostinger PHP API.



Home page with farm-themed mobile layout.



Latihan Pengukuhan topic menu using the available viewport.

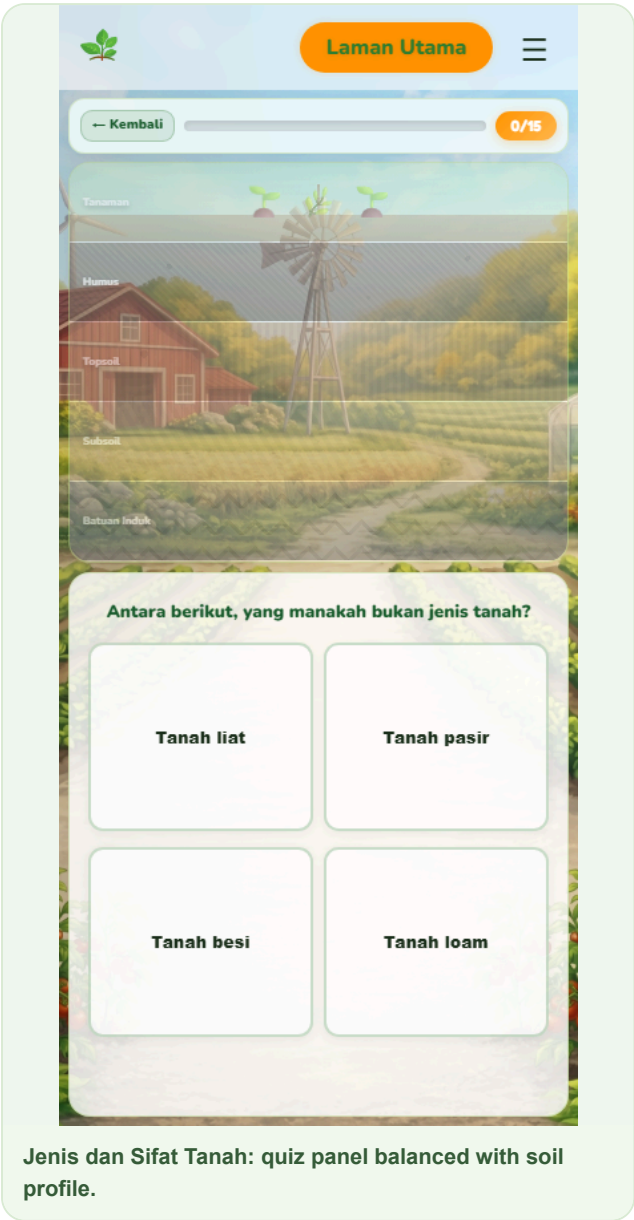


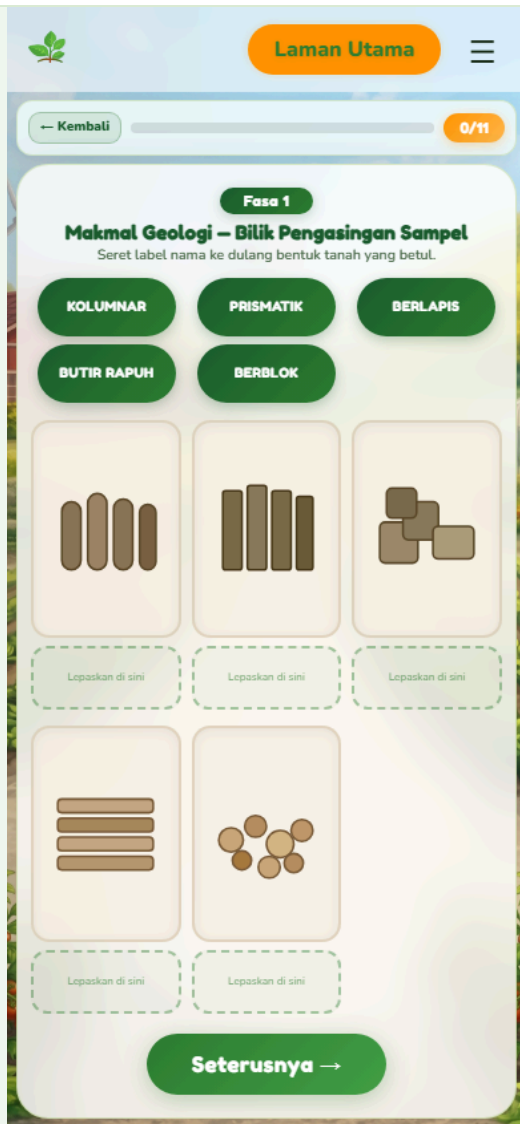
Game locking in 1.3 Baja dan Pembajaan.



Papan Prestasi with five-player leaderboard and top pagination.

8. Game Page Screenshots





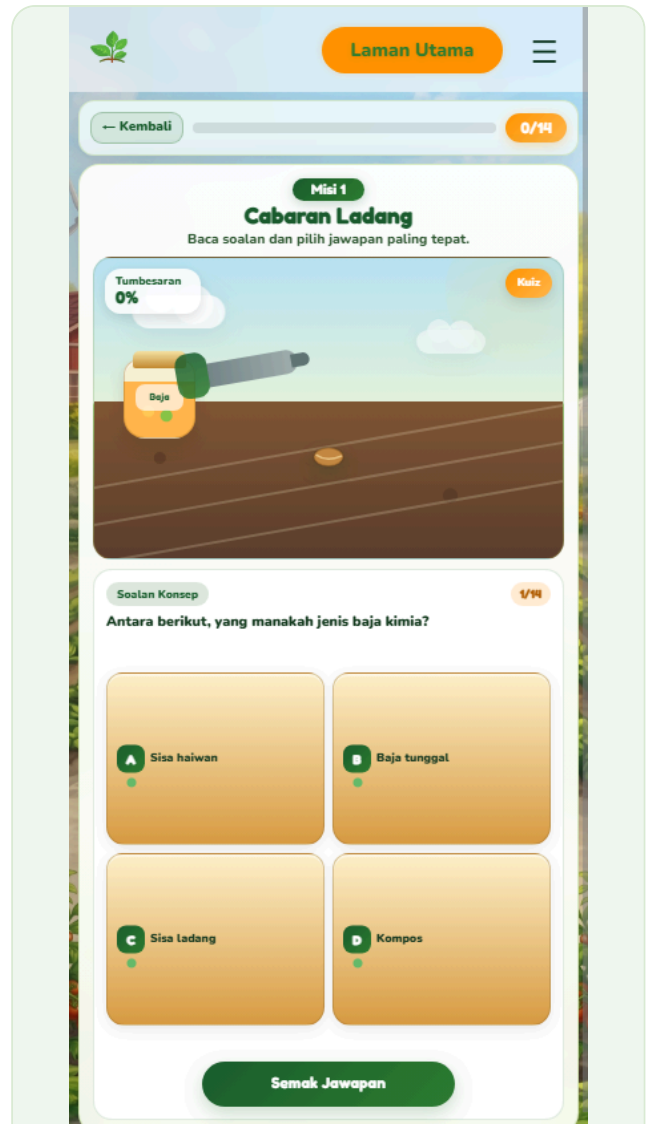
Struktur Tanah: fasa layout resized for one mobile viewport.



pH Tanah: pH scale board expanded to use screen space.



Kaedah Memperbaiki Tanah: answer panels and scene fit inside one viewport.



Jenis Baja: scene, question, and answer cards in a single mobile screen.



Laman Utama

☰

← Kembali

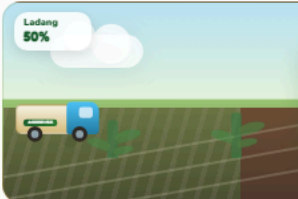
5/10

Misi 6

Pengurus Gudang Baja

Masukkan jumlah kos pada mesin juruwang.

Ladang 50%



PESANAN BELIAN

Item: Baja Urea

Berat: 43.48 kg

Kadar: RM120.00 / 50kg

KIRA PESANAN 1/5

Pesanan Belian

6/10

Diminta: 43.48 kg baja Urea. Harga semasa: RM120.00 bagi 50 kg. Masukkan jumlah kos yang tepat.

Formula Rujukan

RM 0

1

2

3

4

5

6

7

8

9

.

0

Padam

Luluskan

Pengiraan Kos Baja: calculator mission with purchase order and truck scene.

9. Testing and Verification

- Verified page layouts using Chrome mobile emulation at 390 x 844.
- Checked leaderboard display with five visible players per page.
- Confirmed high-score logic uses best score locally and in MySQL.
- Checked the calculator/keypad mission fits in a single phone viewport.
- Checked truck animation in Pengiraan Kos Baja: entry from left and delivery exit to right.
- Ran **git diff --check** on the documentation file; no whitespace errors were reported.

10. Remaining Notes

- Hostinger PHP/MySQL deployment should be tested through the live domain after uploading the latest files.
- The local screenshot for Papan Prestasi uses mock leaderboard rows for presentation because local file preview cannot execute the hosted PHP API.
- Future improvement: add a small admin/export page for downloading leaderboard records as CSV.